

Supplementary materials: Lexical statistics in early noun vocabularies (shape bias project)

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This document supplements the main manuscript *Lexical statistics and the shape bias* (same data and code repository). It provides (S1) kernel-density plots of solidity, count, and shape ratings for **all 16** unilemma languages in Study~1A, **split into two page-sized figures**; (S2) 80% cutoff Venn diagrams, likewise split; (S3) the 4×4 correlation matrix of **word-level shape proportions** across English samples 1–2 and Korean samples 1–2; (S4) the Korean-only age-of-acquisition model reported as a robustness check in the main text; (S5) numeric details of the Study~2B strict- vs. broad-shape coding sensitivity analysis; and (S6) pointers to **browser-ready jsPsych implementations** of the rating tasks (Studies~1A and 2A), described below. **Study~1A Materials coverage** (referenced in the main manuscript): **per-language unilemma counts** and a **semantic-category composition** table and heatmap (linked via `ratings_aoa.csv`); **pairwise Jac-card overlap** between the 16 language-specific noun lists; **participant demographics** from Prolific CSV exports; and **distributions** of per-item rating totals (`totcount`) and response proportions across blocks.

Experiment implementations (Studies 1A and 2A)

The repository folder `experiment files/` (at the project root) contains the exact **HTML + JavaScript** tasks used for online collection. They mirror the procedures described in the main text:

- **Study 1A (English, 16-language unilemma lists):** Open `solidity-ratings.html`, which loads `solidity-ratings.js` (embedded word lists and jsPsych timeline). Participants completed three blocks—solidity, count–mass syntax, and category-organizing feature—in random order (20 nouns per block). On test trials, each *uni_lemma* appears alone as the prompt for solidity and count–mass; the organizing-feature trial asks whether the lemma *belongs to a category that is organized by:* with responses *shape / color / material / none of these*.
- **Study 2A (Korean):** Open `vocab_comp.html`, which loads `words.js` (defines global `words_array`) then `vocab_comp.js`. Instructions follow the clarified wording used in Study 2A; test-trial structure matches Study 1A for the corresponding blocks.

Supporting assets include local copies of jsPsych plugins under `experiment files/jspsych/` and shared styling (`solidity.css`). Opening the HTML files directly from disk (`file://`) works with the scripts as checked in (classic scripts; no ES-module imports required). Serving the folder over a local HTTP server is optional for deployment or debugging.

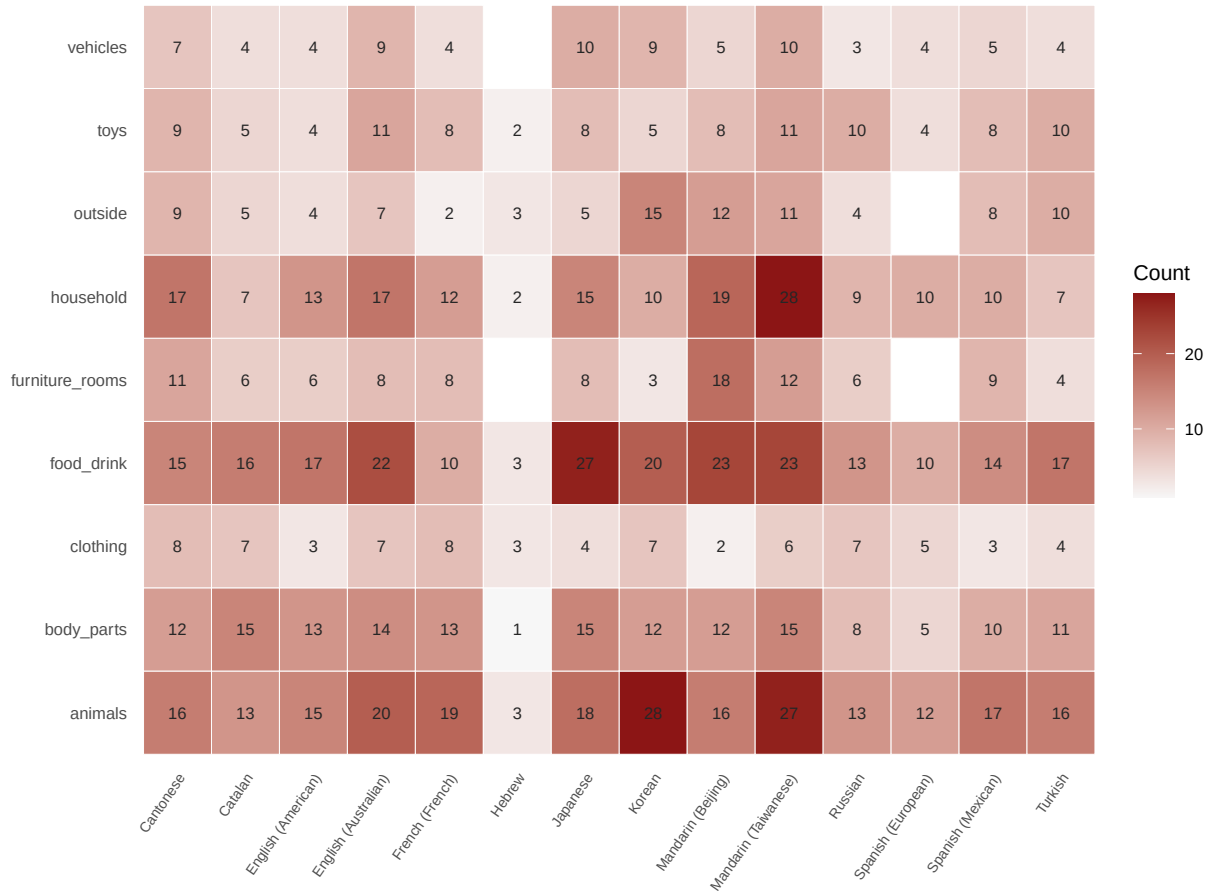


Figure 1: Semantic category counts per language for Study 1A items (categories taken from `ratings_aoa.csv`, aligned with AoA regression syntax). Cell value is the number of unlemmas in each category–language cell.

Table 1: Study~1A: Number of distinct unilemmas rated per Wordbank language list (English-speaking raters).

language	N unilemmas
Cantonese	233
Catalan	257
English (American)	291
English (Australian)	264
French (French)	229
Hebrew	64
Japanese	250
Kiswahili	82
Korean	247
Mandarin (Beijing)	286
Mandarin (Taiwanese)	288
Russian	282
Spanish (European)	227
Spanish (Mexican)	234
Spanish (Peruvian)	53
Turkish	275

Table 2: Study~1A: Same semantic-category counts as the heatmap (wide for

category	Cantonese	Catalan	English (American)	English (Australian)	French (French)	Hebrew	Japanese
animals	16	13	15	20	19	3	18
body_parts	12	15	13	14	13	1	15
clothing	8	7	3	7	8	3	4
food_drink	15	16	17	22	10	3	27
furniture_rooms	11	6	6	8	8	0	8
household	17	7	13	17	12	2	15
outside	9	5	4	7	2	3	5
toys	9	5	4	11	8	2	8
vehicles	7	4	4	9	4	0	10

Note. Categories follow Wordbank semantic coding linked in predictor construction.

Study 1A: Lexicon composition, cross-language overlap, demographics, and rating distributions

Per-language unilemma counts and semantic categories

Pairwise overlap of language-specific unilemma sets (Jaccard index)

Participant demographics (Study 1A; Prolific)

Distributions of rating totals and response proportions (Study 1A)

Study 1A: Sixteen languages (full grids, two panels per plot type)

Cross-sample shape correlations ₃

Korean-only age-of-acquisition model

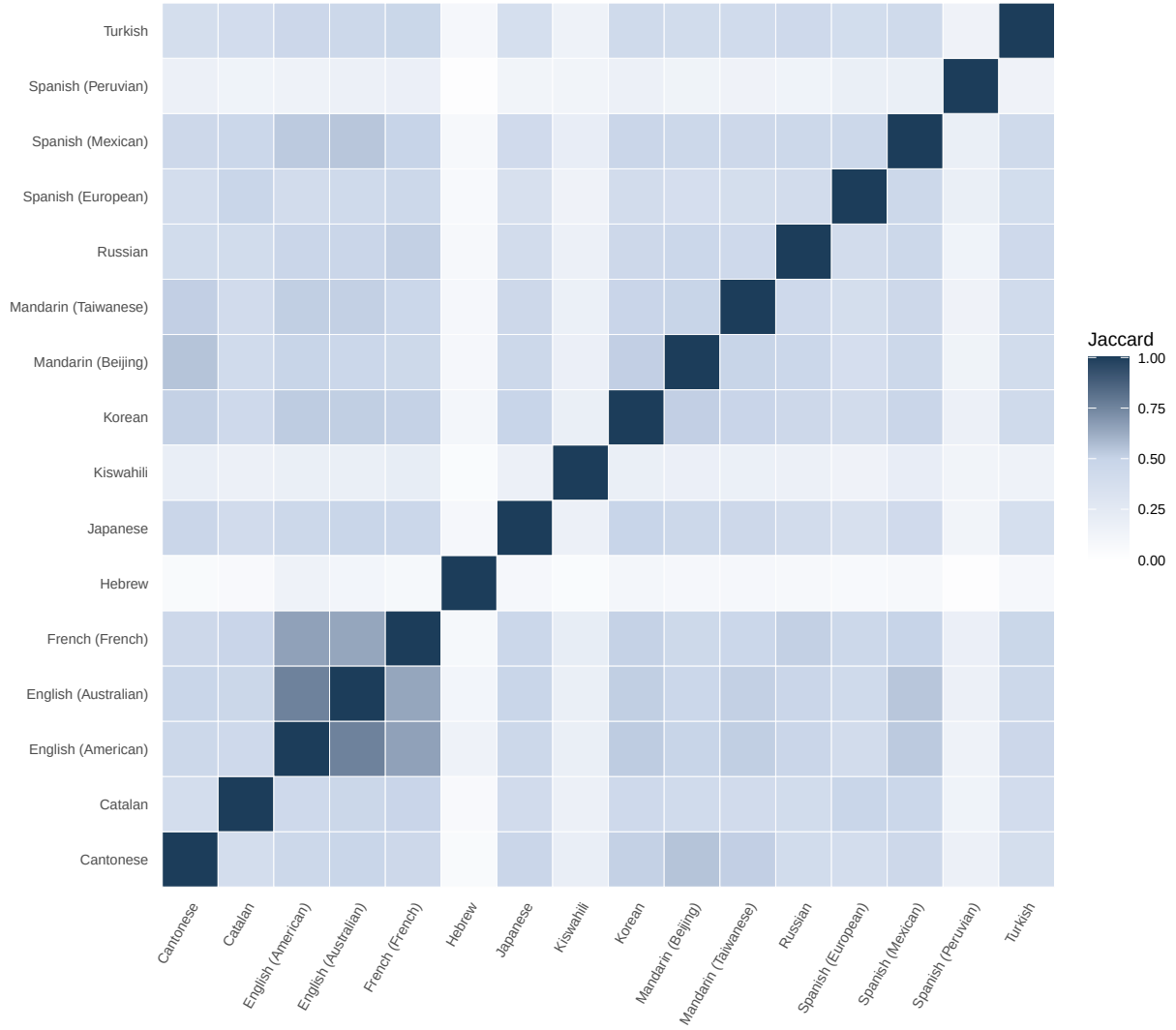


Figure 2: Pairwise Jaccard similarity $|L_i \cap L_j| / |L_i \cup L_j|$ between Study 1A noun lists L_i , defined over distinct `uni_lemma` values per language in `english_sample1_props.csv`. Diagonal = 1.

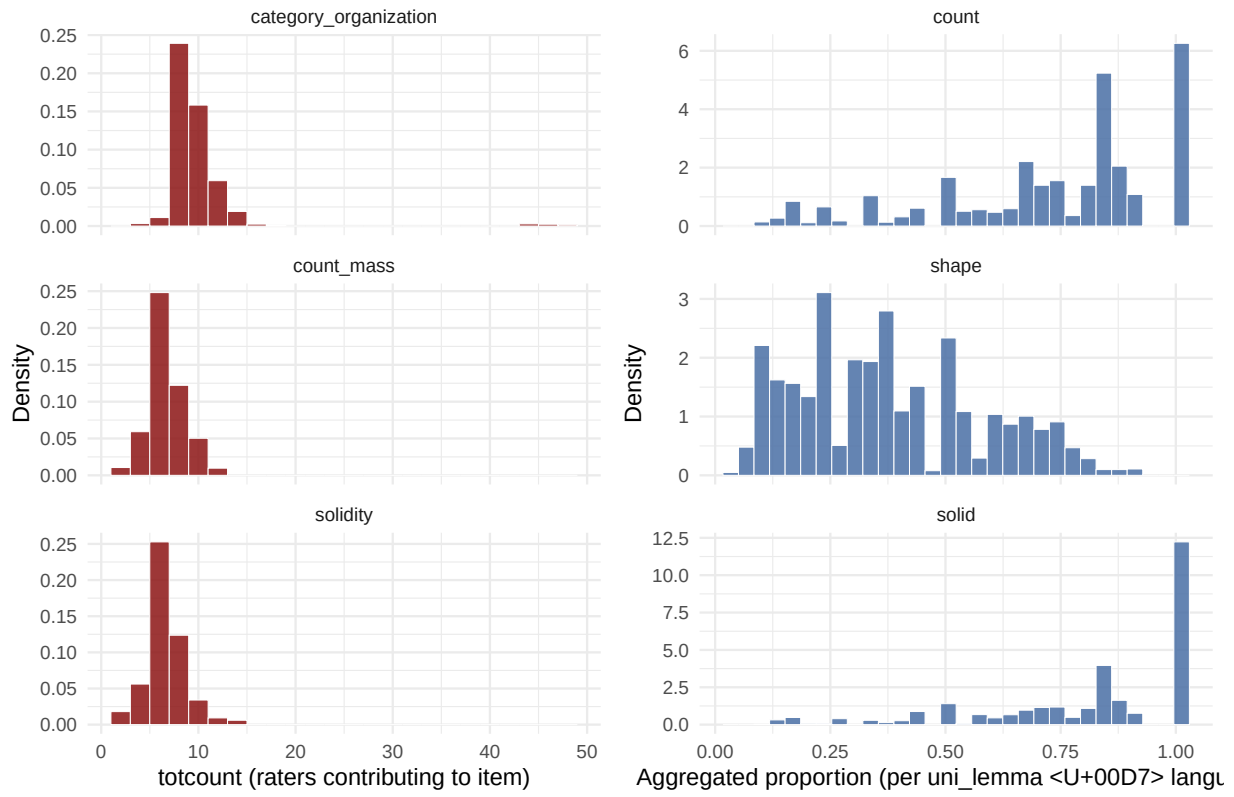


Figure 3: **Left:** Distribution of per-item-language totals (number of participant judgments contributing to each aggregated proportion), by experimental block. **Right:** Distribution of aggregated proportions for solidity, count-noun, and shape responses (coded dimension of proportion column).

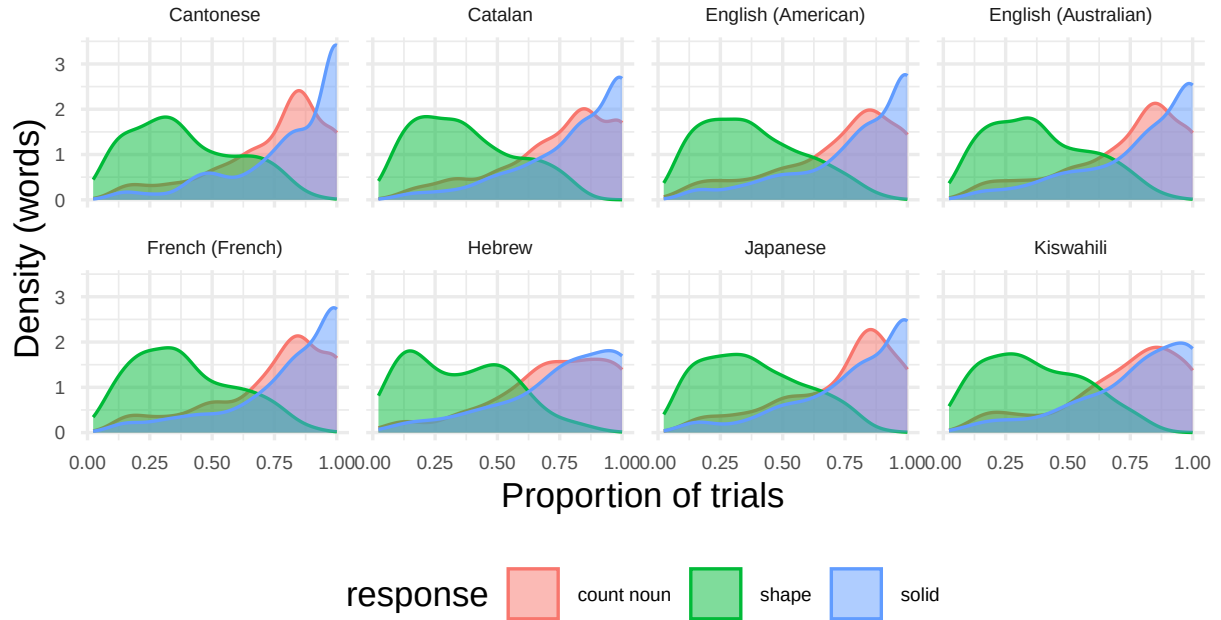


Figure 4: (Part 1 of 2) Kernel density of per-word proportions for solidity, count, and shape responses, Study 1A (eight unilemma languages, alphabetical a–h). See Part 2 for remaining languages.

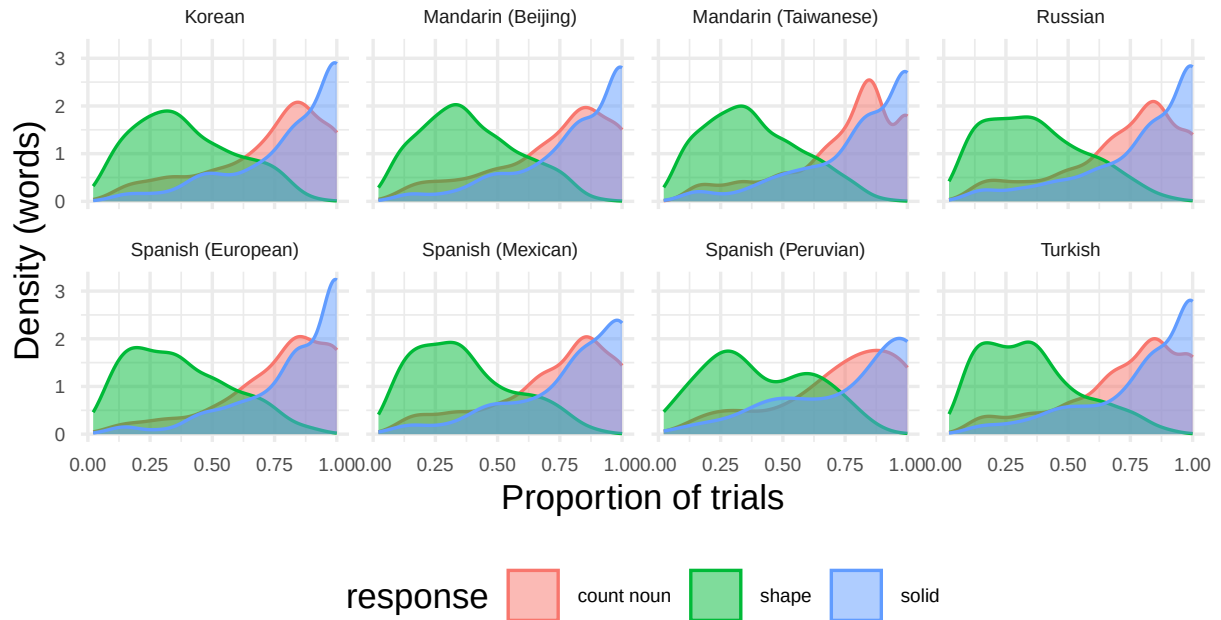


Figure 5: (Part 2 of 2) Same as Part 1 for the remaining eight unilemma languages (alphabetical i–p).

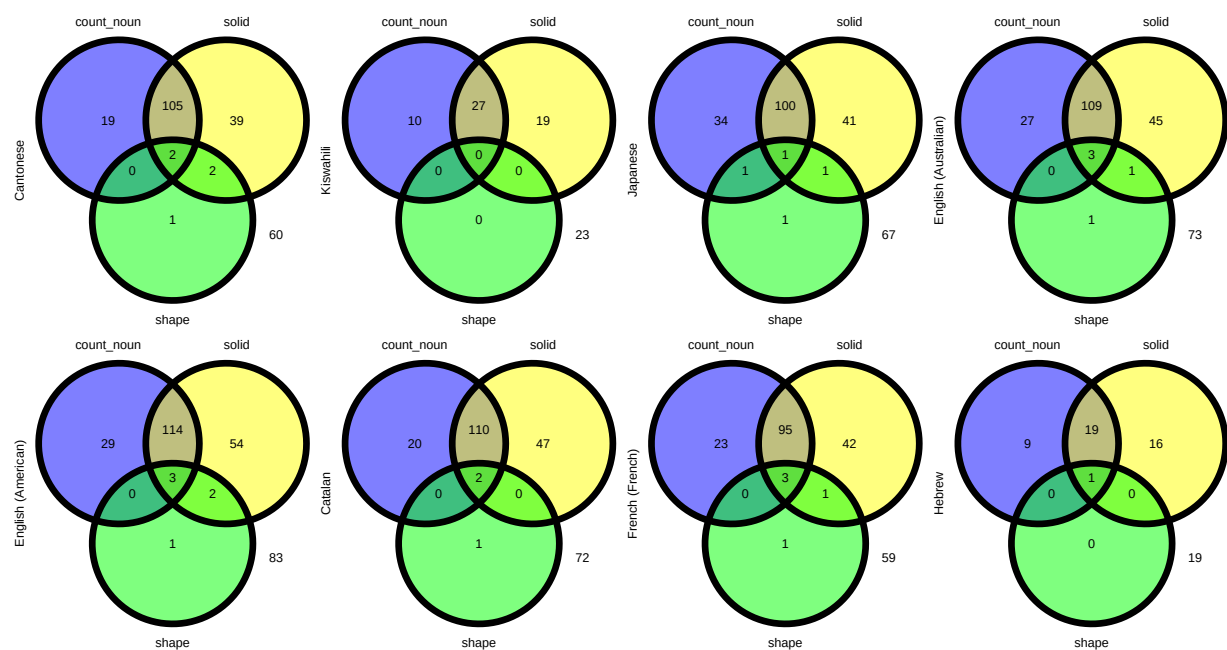


Figure 6: (Part 1 of 2) Venn diagrams (80% cutoff) for solid, count, and shape-organization, Study 1A (eight languages; same ordering as the density figure).

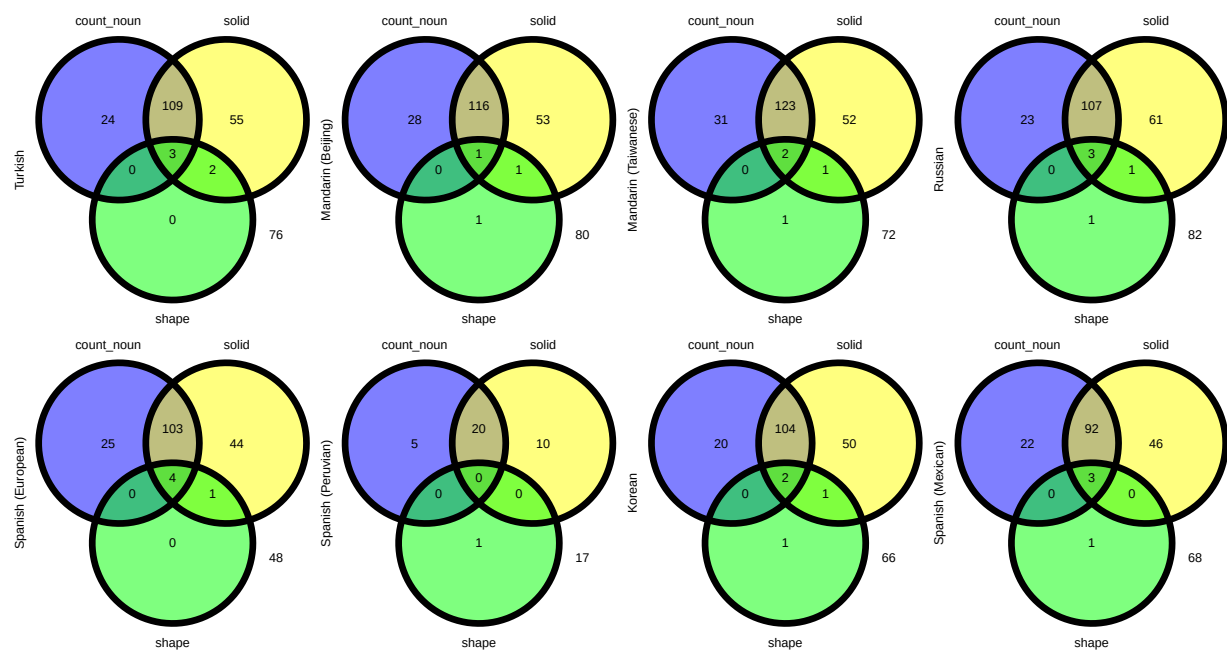


Figure 7: (Part 2 of 2) Venn diagrams for the remaining eight languages.

Table 3: Study~1A: Demographic summary from merged Prolific export files (`\texttt{data/demographics/first-sample-english/}`).

Statistic	Value
N participants (rows in combined Prolific exports)	639
Mean age (years)	41.24
SD age	14.17
Min age	18
Max age	79

Table 4: Study~1A: Self-reported sex (Prolific).

sex	n	pct
CONSENT_REVOKED	60	9.4%
Female	326	51.0%
Male	250	39.1%
Prefer not to say	3	0.5%

Table 5: Pearson correlations of word-level shape-organization proportions across the four norming samples (Study 1A = eng1, Study 1B = eng2, Study 2A = kor1, Study 2B = kor2).

Sample	eng1	eng2	kor1	kor2
eng1	1.00	0.54	0.33	0.25
eng2	0.54	1.00	0.42	0.35
kor1	0.33	0.42	1.00	0.38
kor2	0.25	0.35	0.38	1.00

Note. Lower triangle matches `psych::corr.test` output; diagonal is 1.

`manuscript.Rmd` chunk `kor2-sensitivity` from `data/predictors/korean_sample2_props.csv`. For submission, paste the latest inline values from the rendered main text here if the journal requires a static supplement table.

Table 6: Fixed effects for the Korean-only model predicting age of acquisition (months) from corpus log-frequency, concreteness, and Korean-averaged solidity, shape, and count-noun ratings.

Predictor	Estimate	SE	df	t	p
(Intercept)	27.83	2.81	62.0	9.91	< .001
logfreq	-0.73	0.34	61.3	-2.17	.034
concreteness	-4.45	2.94	62.0	-1.51	.135
mean_solid_kor	5.77	3.19	60.1	1.81	.075
mean_shape_kor	1.02	2.50	51.9	0.41	.686
mean_count_kor	-7.59	2.73	62.0	-2.78	.007

Note. Random intercept for semantic category. Predictors use the same scaling as in the main text where applicable.

Table 7: Sensitivity analysis for Study~2B multiple-selection trials (per main text). $P(\text{shape})$ = item-level shape proportion under each coding rule.

Description	Value
Organizing block trials (total, weighted N)	7,408
“Shape only” (strict)	4,323
Shape + other feature (broad)	2,170
Mean $P(\text{shape})$, strict	0.584
SD, strict	0.161
Mean $P(\text{shape})$, broad	0.877
SD, broad	0.116